

INFORMATICS INSTITUTE OF TECHNOLOGY

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In Collaboration with

UNIVERSITY OF WESTMINSTER

Game Engine Architecture – Game Design Document - Coursework

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GAME OVERVIEW

Working project title: Abandoned Outpost

Target Audience: The main target audience of the game is teenagers or young adults who like exploring a map or in general enjoy the scenery of a map.

Features/USPs:

There is no time limit for the user to complete the game. Any amount of time can be taken by the gamer to explore the map and enjoy the scenery.

The only way to end the game is by reaching a certain area. This feature adds to the enjoyment of the game where the gamer could search for the exit point if the user wants to close the game.

There are a couple of interactive objects in the game and one of those interactive objects contributes to the main plot of the game.

ORIGINAL GAME DESIGN

Game Concept:

Originally, the game was envisioned as a straightforward first-person shooter (FPS) centered around the player's task of clearing a military outpost and retrieving specific documents to achieve the game's objective. Due to certain complexities that cost time, the FPS shooter concept was abandoned in favour of a different approach but still meet the requirements of the coursework. The revised game required players to explore their surroundings and retrieve documents to complete the game.

The game draws inspiration from military operations, particularly the Sri Lankan Civil War, which occurred between July 23, 1983, and May 18, 2009. The specific level and storyline within the game are based on this conflict, following the perspective of a Sri Lankan Special Forces soldier engaged in a covert operation. The mission involves clearing an outpost and retrieving vital documents. This operation is considered a tactical sub-operation, forming part of a broader military campaign known as Operation Liberation, also referred to as the Vadamarachchi Operation. Operation Liberation was a military offensive executed by the Sri Lankan Armed Forces. It is important to note that this sub-operation is entirely fictional and not based on real events.

Given the change in gameplay mechanics, the game was aptly named "Abandoned Outpost." Players assume the role of a Special Forces soldier who parachutes into the designated area and embarks on a mission to locate and retrieve the documents, all while finding an escape route to successfully complete the game.

The concept behind "Abandoned Outpost" draws significant inspiration from popular video game series such as Call of Duty, particularly the earlier installments like Call of Duty 4: Modern Warfare (2007), Call of Duty: World at War (2008), and Call of Duty: Black Ops (2010). By reimagining the initial FPS concept, the game offers an engaging and immersive experience that incorporates elements of military operations, strategic retrieval, and the spirit of classic first-person shooter games.




'War Pig'
Day 3 - 16:30:10
Sgt. Paul Jackson
1st Force Recon Co., U.S.M.C.



Setting and Environments:

The game's setting is situated in a mountainous region in the northern part of Sri Lanka, although it should be noted that most of the game's content is fictional. The actual operation that inspired the game primarily took place in the north region, characterized by a terrain that is predominantly low and flat, transitioning into rolling plains with mountains located in the south-central interior.

Within the game's environment, players will encounter a diverse landscape. The predominant features include lush green forests, except for the road leading to the outpost, which is depicted as a dirt path. The outpost itself is comprised of typical bunkers, a main house, and various vehicles. Additionally, players can explore a water body within the game, adding an additional element to the environment. The water lake system features advanced fluid surface rendering with realistic environmental reflections showing the sky and surrounding landscape. The water demonstrates proper depth rendering, transparency effects, and underwater material transitions, positioned strategically within a natural terrain depression to enhance both visual appeal and environmental storytelling. As players venture further up into the mountains, they will be greeted by an increased presence of trees, plants, and the captivating sight of a beautiful sunset. Additionally, helicopters may be encountered in this elevated area, adding a dynamic aspect to the gameplay.

The design and layout of this game map draw inspiration from various sources, with a primary influence being nature-focused maps found in Call of Duty and similar games. These references have been adapted and integrated into the game to create a visually appealing and engaging environment for players to explore and navigate.







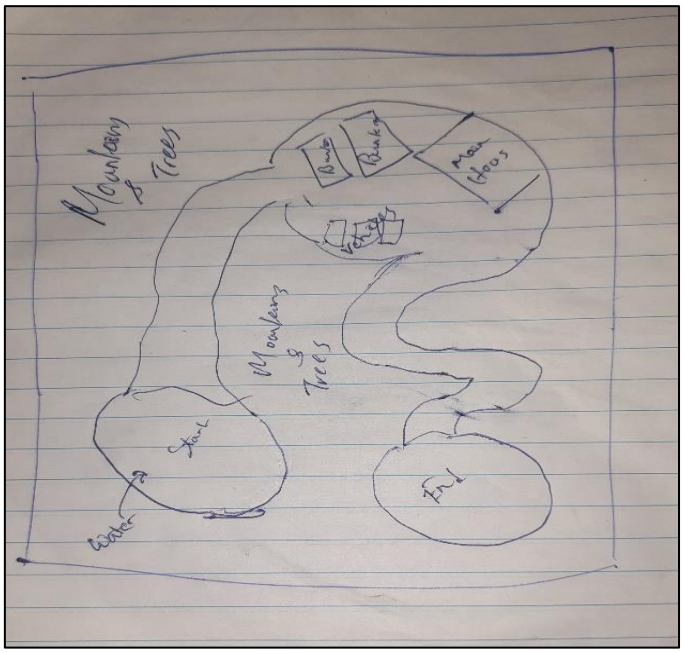


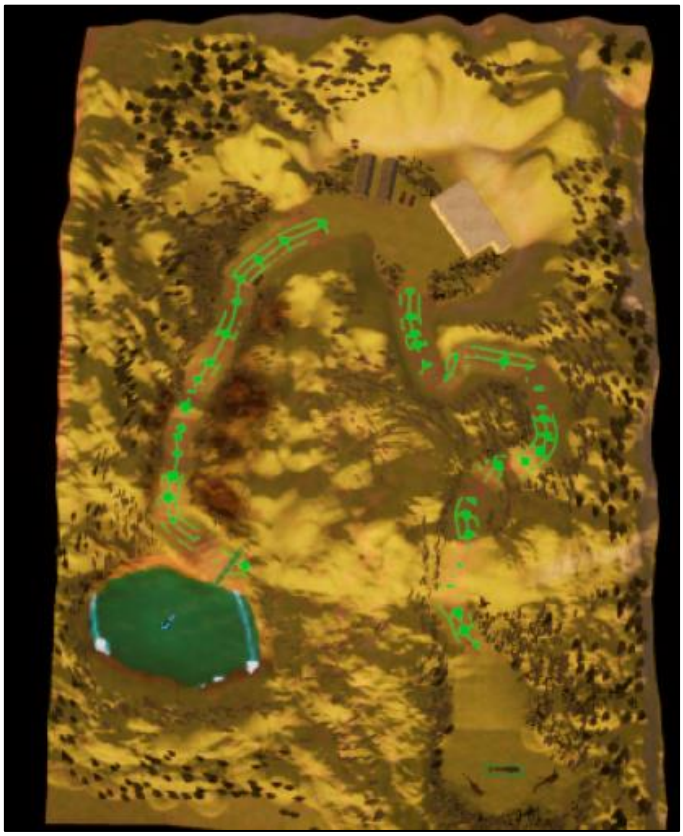




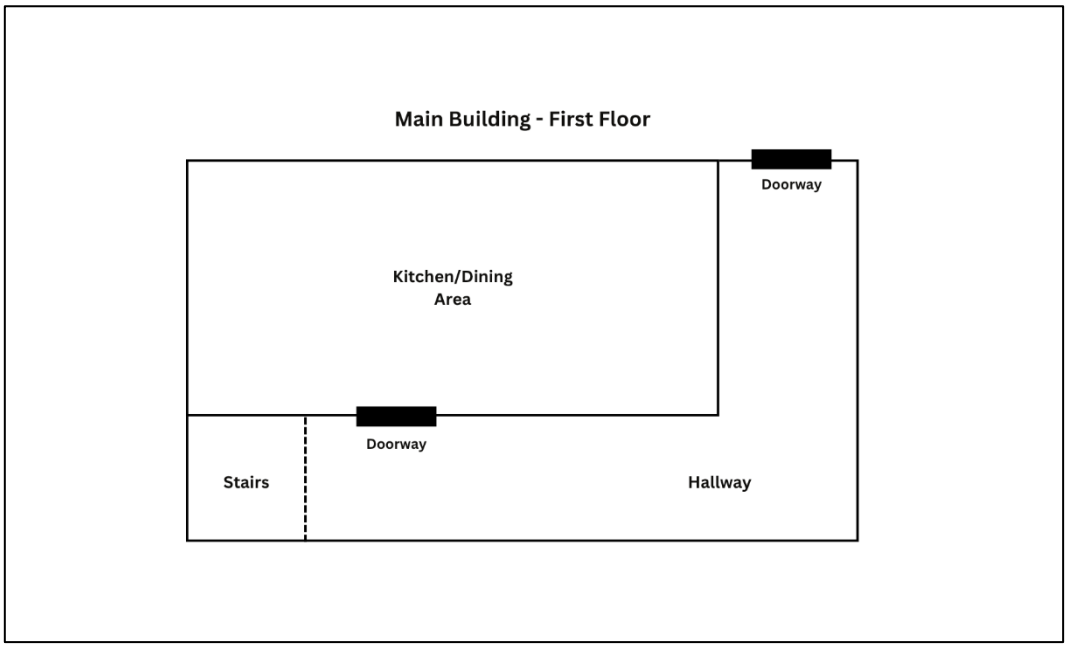


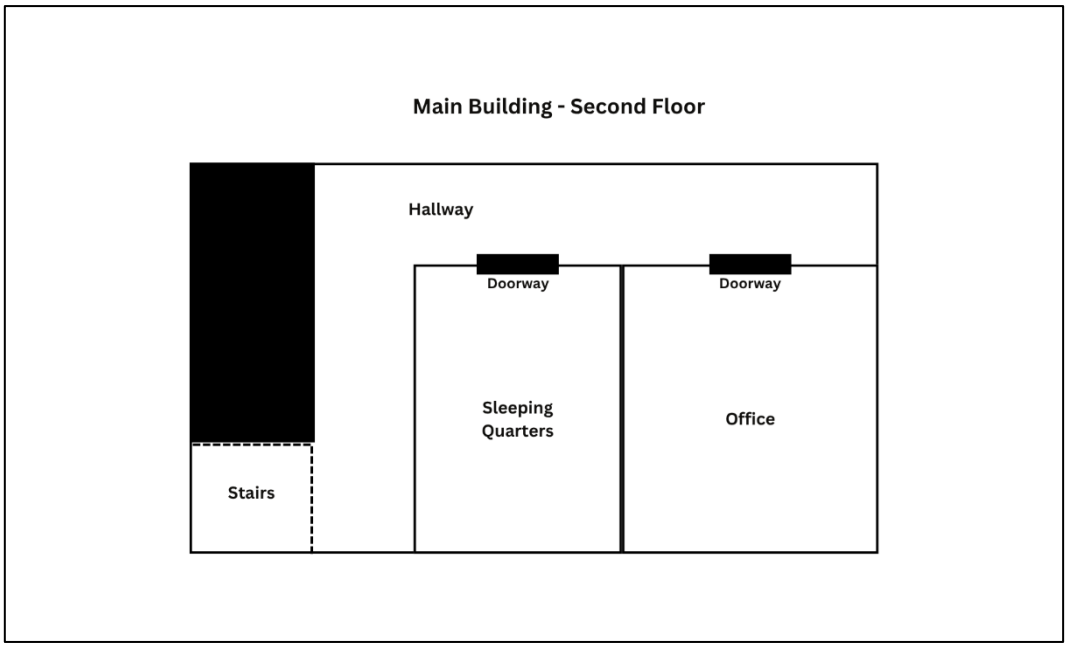
Indeed, by incorporating references from various sources, particularly drawing inspiration from nature-focused maps in games like Call of Duty a well-structured and captivating game environment was created. The unattractive hand drawn layout was the initial layout that ran through the head, so it was jotted down accordingly.

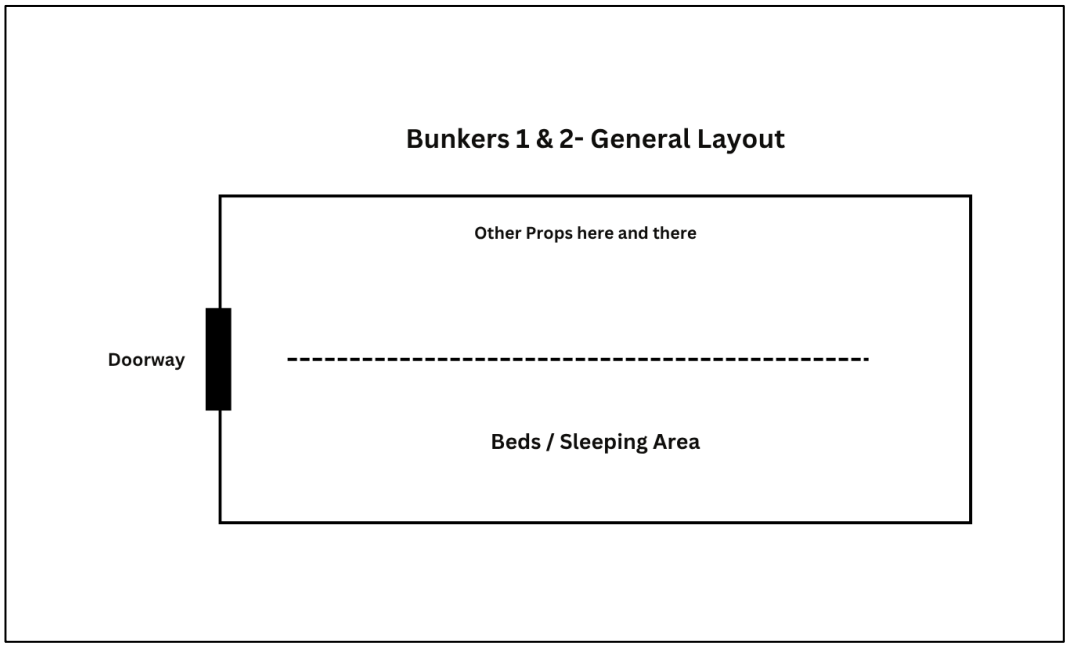




A general layout of the buildings of the game map was considered as follows:







Key Gameplay features:

"Abandoned Outpost" offers a range of key gameplay features that contribute to an immersive and captivating experience. The game excels in several aspects, including intuitive and effective controls, immersive audio design, and visually captivating scenery.

The controls and mechanisms in "Abandoned Outpost" have been meticulously designed to provide players with a simple yet effective means of navigating the game world and interacting with objects. It strikes a balance between functionality and ease of use, ensuring that players can seamlessly engage with the game without feeling overwhelmed or confused by complex controls. The control scheme of the game has been carefully crafted to be intuitive and user-friendly. Players will find that the controls respond smoothly and accurately to their inputs, allowing them to move their character effortlessly throughout the game world. Whether it's walking, running, or manoeuvring through tight spaces, the controls offer a high level of responsiveness, providing a satisfying and immersive gameplay experience.

The game features a unified ‘E’ key interaction system that serves as the primary mechanism for all interactive elements. Players receive clear visual prompts when approaching interactive objects, ensuring intuitive gameplay flow. Specific implementations include:

Document Collection System: Players can collect crucial documents using the E key, which triggers immediate UI feedback displaying "Document Retrieved" along with audio confirmation. The document object disappears upon collection, providing clear progress indication.

Interactive Furniture: The game includes an interactive drawer system featuring smooth open/close animations with realistic sound effects. Players can repeatedly interact with drawers using the same E key system, demonstrating advanced Blueprint functionality.

These interaction mechanisms utilize Blueprint visual scripting with integrated sound effects, UI feedback systems, and state management to create a seamless and responsive gameplay experience.

The audio design in "Abandoned Outpost" is carefully crafted to enhance immersion and create a captivating gameplay experience. The game features thoughtfully selected music that complements the theme and setting, providing a soothing and atmospheric backdrop without overpowering the player. Immersive background sounds, such as birds chirping and wind rustling, add realism and depth to the game world, while sound effects offer valuable feedback and cues during interactions and game progression. The audio elements work together to create a vibrant and engaging auditory experience, making the game a truly immersive and enjoyable game.

"Abandoned Outpost" captivates players with its visually stunning scenery. The game's lush greenery, water elements, and breathtaking sunsets create a serene and inviting atmosphere that draws players into its immersive world. The attention to detail in the environment is evident, enhancing player engagement and allowing them to truly appreciate the beauty of the virtual landscape. From the swaying trees to the tranquil waters, every element is meticulously crafted to create a visually captivating experience.

The visual aesthetics of "Abandoned Outpost" play a significant role in setting the tone and mood of the game. The serene and picturesque surroundings create a sense of tranquillity and exploration, encouraging players to immerse themselves in the game world. The attention to detail in the environment design brings the virtual world to life, making it feel vibrant and realistic.

The combination of these gameplay features creates an immersive and calm experience in "Abandoned Outpost." The smooth controls, immersive audio, and visually captivating scenery work together to transport players into a state of tranquillity and captivation. Players can explore, discover, and engage with the game world at their own pace, providing an opportunity to unwind and enjoy the serene virtual environment.

In summary, "Abandoned Outpost" offers a well-rounded gameplay experience, providing intuitive controls, immersive audio design, and visually captivating scenery. It creates an immersive and calm atmosphere that allows players to dive into a tranquil virtual world and enjoy the serenity it offers.

Game Rules:

In "Abandoned Outpost," the game begins with the player parachuting into the game world, marking the start of their adventure. While the game provides instructions, players have the freedom to choose their own path and explore at their own pace, as there is no imposed time limit. The game concludes when the player reaches a specific designated area, signifying the end of their mission.

Players can interact with specific objects within the game world. While not every object is interactive, there are certain items that players can engage with to progress through the game or discover additional information. These interactive objects may include doors, switches, drawers, or other elements that contribute to the overall exploration and puzzle-solving aspects of the game.

On the other hand, vehicles within the game serve a different purpose. Rather than being fully usable by the player, they are primarily designed as props to enhance the realism and immersion of the game world. These vehicles contribute to the visual aesthetics and setting of the abandoned outpost, creating a believable environment for players to immerse themselves in. While players may not be able to drive or operate the vehicles, their presence adds to the atmosphere and helps to establish the context and narrative of the game.

By incorporating these rules and mechanics, the game ensures a balanced and coherent gameplay experience. The interactive objects provide opportunities for players to engage with the environment and progress through the game, while the presence of non-interactive objects and props, such as vehicles, adds depth and immersion to the game world. This combination allows players to navigate and interact with the game world in a meaningful way, contributing to an enjoyable and cohesive gameplay experience.

Overall, the game offers a sense of freedom and exploration, allowing players to make choices and progress through the game at their own pace. The objectives and limitations within the game provide structure while still offering room for player agency and immersion.

Game Progress:

The game begins with the player accessing the main menu and starting the game. Upon starting, the character is introduced by parachuting into the designated level, which serves as the starting point of the gameplay. Once landed, the game world is unlocked, and the player is granted the freedom to explore and progress at their own pace.

While the game provides instructions to guide players, they also have the option to deviate from the given directions and explore the environment independently. This freedom allows players to fully immerse themselves in the game, discover hidden areas, and uncover additional clues or documents that contribute to the overall narrative. The game's objectives revolve around locating and retrieving crucial documents, which serve as milestones in the player's progression. However, there is no time limit or strict linear path to follow. Players can take their time, engage with interactive objects, and interact with the environment at their own pace.

By providing this balance of freedom and guided progression, "Abandoned Outpost" offers a dynamic and engaging gameplay experience. Players have the opportunity to exercise their agency and make choices that impact their exploration and success in the game. Whether following instructions or venturing off the beaten path, players can unlock the full potential of the game and experience its immersive world.

Game Flow:

In the game flow of "Abandoned Outpost," the overall design aims to provide a smooth and intuitive experience for players, ensuring they encounter minimal complications or obstacles while progressing through the game level. However, one significant challenge that emerged during the game development was related to the deactivation of the final trigger box.

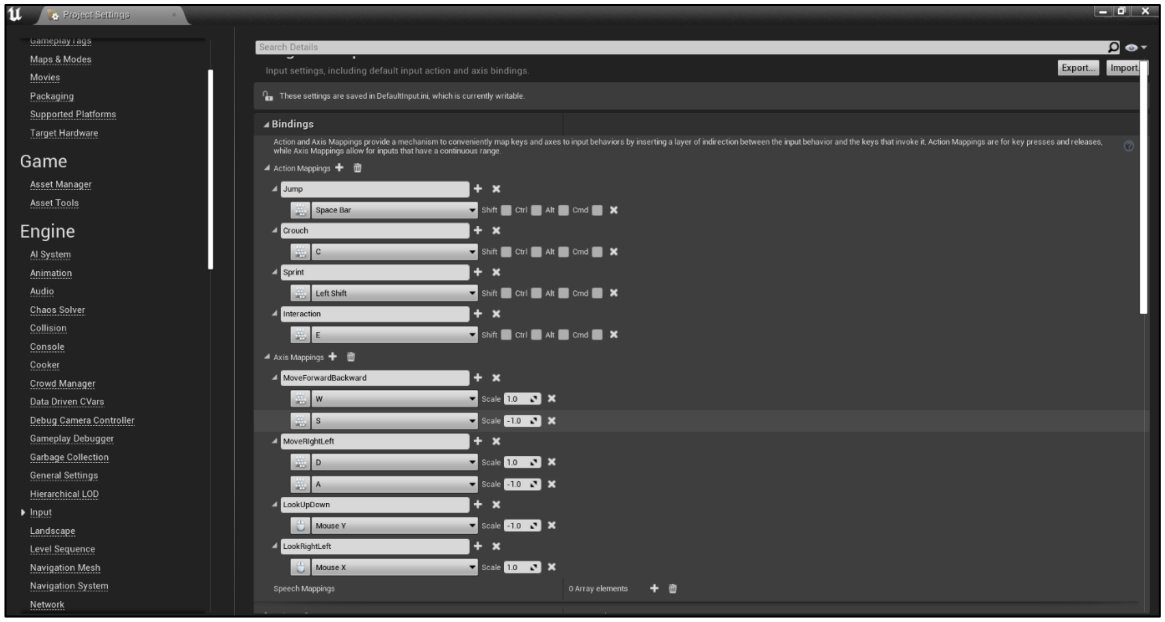
Ideally, the intention was for the game to end and a widget to pop up, triggering the conclusion of the game, only when the player had successfully located and retrieved the documents. However, due to technical limitations or difficulties in implementing this specific condition, it became a challenge to modify the game flow accordingly. As a result, a decision was made to omit the requirement of having the documents before reaching the designated trigger box. This adjustment was made to ensure a smoother gameplay experience and to avoid potential frustrations for players who may have missed or overlooked the documents.

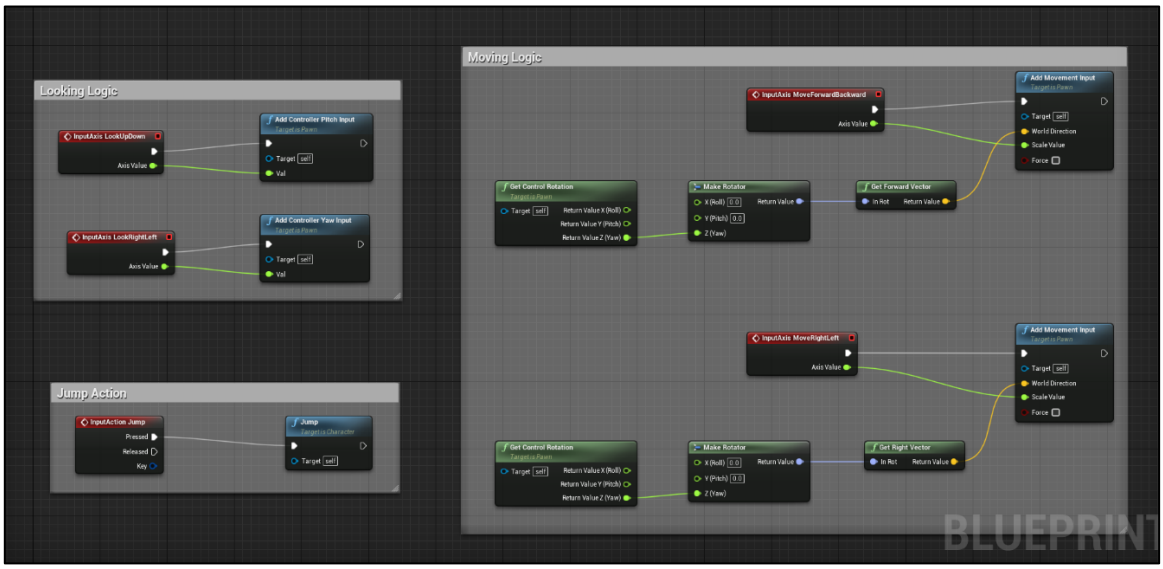
While this adjustment may be seen as a flaw or compromise in terms of the original game design, it was necessary to maintain a balanced and enjoyable game flow. It allows players to progress through the game level without getting stuck or encountering unnecessary barriers, ultimately ensuring a more seamless and engaging experience.

By addressing this challenge and making the necessary adjustments, "Abandoned Outpost" maintains its intended flow and offers players a rewarding and fulfilling gameplay journey.

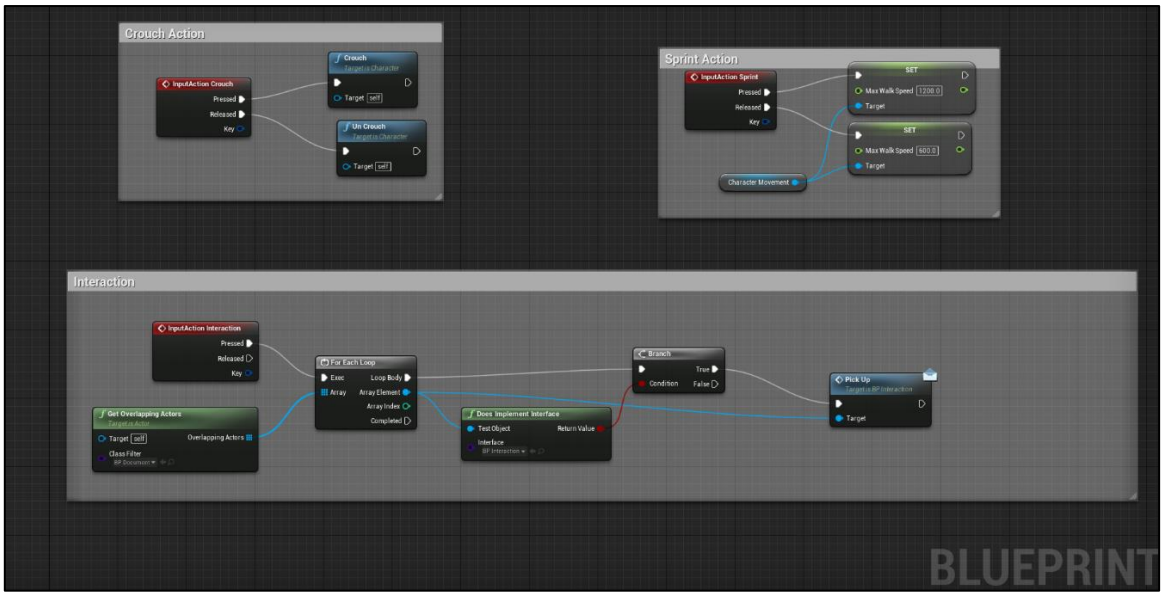
Game Controls:

Players can expect typical key mappings and functions that are commonly found in games of this genre. These include essential controls such as looking around, walking, jumping, sprinting, and crouching. These movement controls allow players to navigate the game world and explore their surroundings.





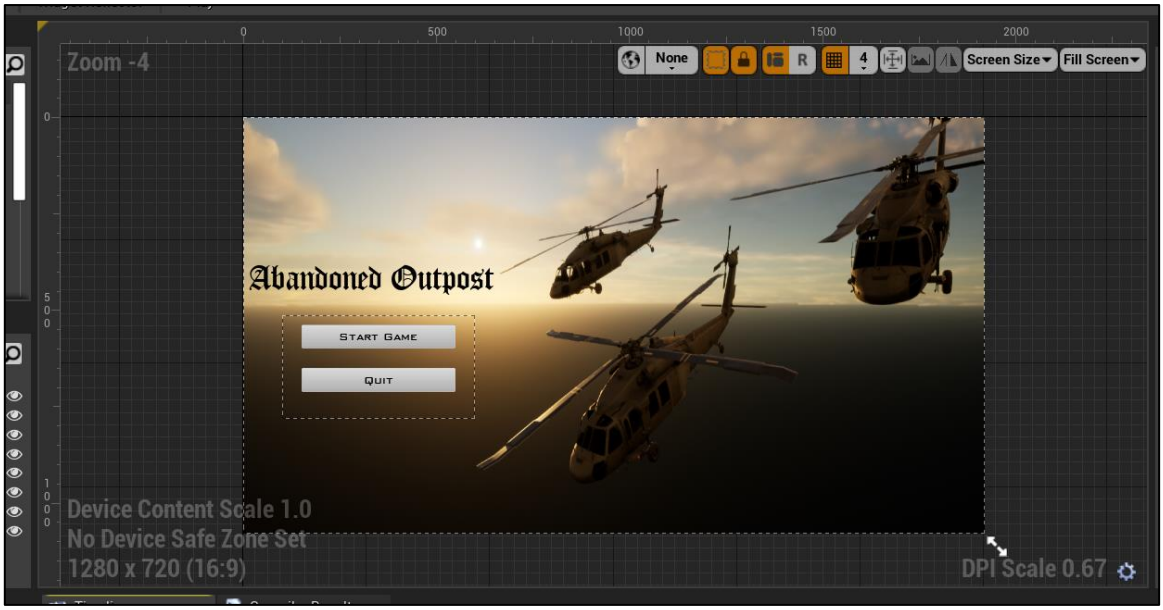
Additionally, interaction controls play a crucial role in the gameplay. Players will be able to interact with various objects and elements within the game environment. This may include picking up objects, opening doors, activating switches, or interacting with specific items necessary for progressing in the game.

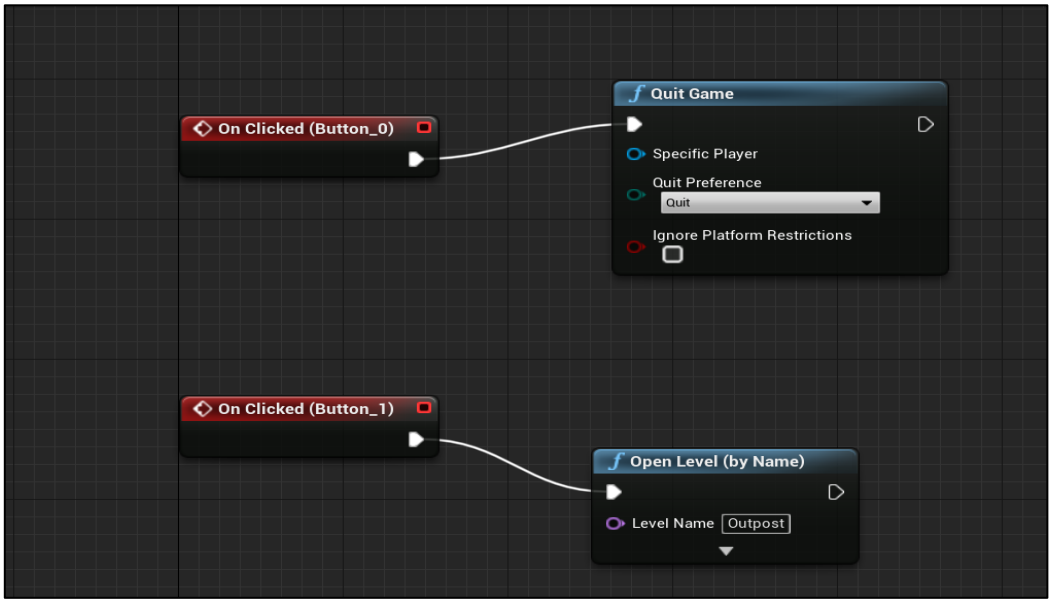


The main menu features a simple and intuitive design, with buttons that provide basic functionality for the player. These buttons typically include options such as "Start Game," and "Quit."

The "Start Game" button allows the player to begin a new game and embark on their adventure in the abandoned outpost.

The "Quit" button allows the player to exit the game and return to the desktop or main menu of their gaming platform.





The menu's layout and design are kept simple to provide a straightforward and user-friendly experience. Players can easily navigate through the menu and access the desired options with ease, ensuring a smooth transition from the main menu into the immersive world of "Abandoned Outpost."

Objects/Characters:

The main playable character takes on the role of a soldier. The player assumes control of this character throughout the game, guiding their actions and decisions. The character serves as the primary vessel for the player's exploration and interaction with the game world. As the main character, the player has various abilities and actions at their disposal. They can engage in basic movements such as walking, running, jumping, sprinting, and crouching, allowing them to navigate the environment with agility. These controls provide the player with a sense of agency and freedom as they explore the abandoned outpost.

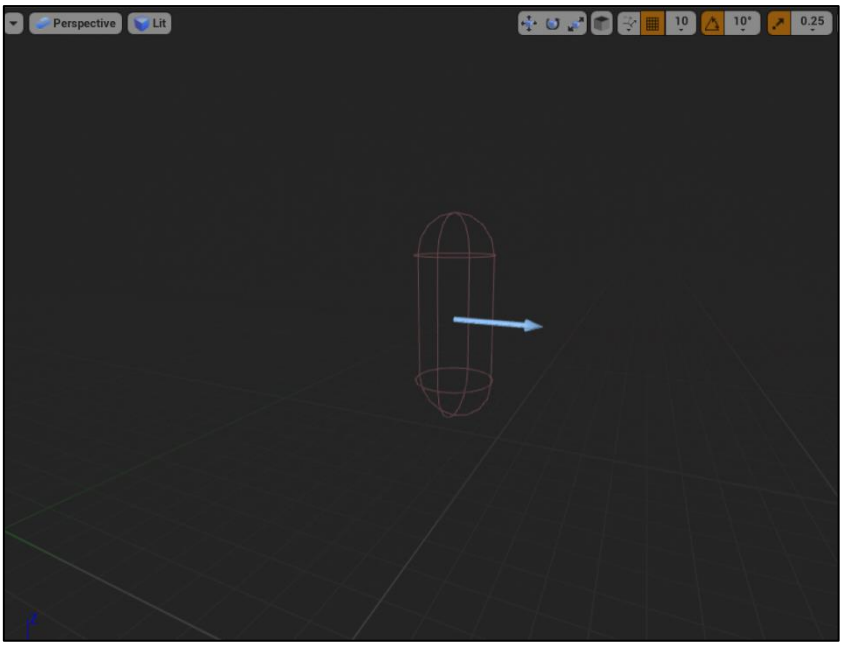
Additionally, the interactive objects within the game serve as significant characters/objects. These objects are carefully placed within the game world to provide opportunities for interaction and progression. These objects play a crucial role in the gameplay, requiring the player's attention and engagement to advance through the game.

Document Objects: Strategically placed collection items that provide immediate player feedback through UI text display and audio cues upon interaction.

Furniture Systems: Interactive drawer mechanisms demonstrating advanced Blueprint functionality with animation timelines, sound integration, and repeatable interaction states.

These objects utilize the unified E key interaction system, ensuring consistent player experience and demonstrating sophisticated Blueprint implementation with proper state management and feedback systems.

The main character and interactive objects work together to create a dynamic and immersive experience for the player. The player's control over the main character, coupled with their interactions with the objects, drives the gameplay forward and allows for meaningful exploration and discovery within the abandoned outpost.






Click for Mouse Control



Game assets management:

Managing game assets through an asset pipeline is a crucial aspect involves several steps to ensure efficient organization, importing, and referencing of different types of assets.

The first step is pre-conditioning the assets before importing them into the game project. This may involve preparing the assets in external software to ensure they meet the required specifications and quality standards. Preconditioning was not required during the making of this game. Once the assets are pre-conditioned, they need to be organized in a structured folder hierarchy within the game project. This helps maintain a clear and organized asset library, making it easier to locate and manage assets during development. Categorizing assets based on their type, such as textures, audio, or animations, can further streamline the organization process. This was done and must be done from the beginning to keep the assets organized.

The next step is importing the assets into the game project using the appropriate tools or import functions provided by the game engine or development environment. This involves specifying the correct settings for each asset, such as compression settings for textures or import options for 3D models. Properly configuring these settings ensures that the assets are imported in the desired format and quality level, optimizing performance, and maintaining visual or auditory fidelity.

When working with assets from external sources or packages, it is essential to maintain proper referencing. This involves establishing and maintaining a clear link between the imported assets and their corresponding locations within the game project. It is also crucial to update asset references if the assets are moved or renamed within the project to ensure consistency and prevent errors. There were times during the making of the project when some assets were not properly named or imported which caused a lot of errors and disturbances.

External assets maintain clear source attribution through organized folder structures, preserving original package names for proper referencing compliance. Strategic placement differentiates between gameplay-critical assets (document collection, interactive furniture) and atmospheric world-building elements (vegetation, vehicles, environmental props).

External Assets	Usage
GrassLandscape (Marketplace)	Terrain & Landscape Materials and Textures
MS_ConstructMudV1 (Megascans)	
MS_DebrisV2 (Megascans)	
MSPresets (Megascans)	

BlackAlder (Marketplace)	Vegetation & Foliage
FL_Scrub (Marketplace)	
FL_WetFlatwoods (Marketplace)	
MS_MilitaryEquipment	Furniture, Vehicles, Military Equipment and Other Props
VigilanteContent (Marketplace)	
FreeFurniturePack (Marketplace)	
MS_BurntPaperDebris (Megascans)	
OldWest (Marketplace)	
StarterContent (Unreal Engine)	

By following a systematic approach to asset management through the asset pipeline, developers can maintain a well-organized and optimized game project. This facilitates efficient workflows, reduces potential issues caused by asset mismanagement, and ensures smooth integration of assets into the final game product.

Walk through:

As the player begins their gameplay experience in "Abandoned Outpost," they are greeted by a captivating main menu that sets the tone for the adventure ahead. Upon starting the game, the player's character is introduced by parachuting into the designated level, creating a thrilling and immersive entry point.

Once on the ground, the player is free to explore the vast and scenic game world at their own pace. They can take in the lush greenery, the serene water elements, and the breathtaking sunsets that paint the horizon. The visual aesthetics of the game create a tranquil and inviting atmosphere, enticing the player to embark on their journey.

As the player moves through the game world, they encounter various interactive objects that add depth to their exploration. Some objects can be interacted with, allowing the player to pick them up, and open objects. Interactive objects allow players to use the ‘E’ key interaction system to collect documents (with immediate "Document Retrieved" UI feedback and audio confirmation) and operate furniture like drawers (featuring smooth animations and sound effects). These interactions contribute to objective completion and progression within the game.

Throughout the gameplay experience, the player is accompanied by an immersive audio design. Soothing music plays in the background, enhancing the atmosphere and creating a sense of calm. The subtle sounds of birds chirping and the wind rustling through trees further immerse the player in the virtual environment.

The objective of the game is to locate and retrieve crucial documents while navigating the challenges presented. The player must carefully explore the abandoned outpost, searching for clues and hints to guide their progress. There is no time limit, allowing the player to take their time, soak in the surroundings, and approach the gameplay with a relaxed mindset.

As the player successfully retrieves the documents and reaches the designated area, the game concludes, providing a satisfying sense of accomplishment. The overall gameplay experience in "Abandoned Outpost" is designed to be immersive, serene, and engaging, offering players a tranquil and captivating adventure within a beautifully crafted virtual world.